TRENVO – MUSIC TREND PREDICTOR

One-Page Project Abstract | Prepared by Tannu Verma

Abstract

Trenvo – Music Trend Predictor is a music data analysis and trend prediction platform designed to help users understand what is trending in the music industry and identify songs and artists that may gain popularity in the future. The system brings together music engagement indicators such as popularity, streams, views, searches, likes, release information and social-media activity to provide an easy-to-understand view of current music trends. The proposed platform is intended to combine data collection, preprocessing, analysis, visualization and machine-learning-based prediction within a simple Streamlit interface.

Key Features

Trenvo provides a real-time-style music dashboard with key performance cards such as **Total Songs, Trending Songs, Top Artist and Top Genre**. It supports trending-song rankings, artist analysis, genre analysis, song and artist search, and filters based on genre, artist, release year and popularity. Users can compare artists using engagement, growth and trending-song metrics. Weekly trend tracking visualizes how rankings rise, peak and decline over time. An additional “Ask Why It's Trending” concept can provide a simple explanation of the factors associated with a song's current trend.

Technology & Methodology

The project uses **Python, Pandas and NumPy** for data collection, cleaning and preprocessing, while **Matplotlib/Seaborn** can be used for analytical visualizations. **Scikit-learn** is used for machine-learning-based trend prediction. **Streamlit** provides the beginner-friendly web interface. Data may be obtained from CSV datasets or APIs such as Spotify and YouTube, together with search-trend and social-engagement data. The workflow consists of collecting raw data, removing duplicates and handling missing values, standardizing fields, engineering a trend score, analyzing songs/artists/genres, visualizing results and applying a classification or scoring model to estimate future trending potential.

Prediction Approach

Historical music records are transformed into useful predictive features such as popularity, stream/view counts, engagement, release information, search activity and growth-related measures. A suitable Scikit-learn model can then learn the relationship between these historical indicators and a defined trending outcome. The model produces a prediction or trend score that can be used to categorize a song or artist as having lower or higher potential to become trending. The prediction is intended as an analytical estimate rather than a guarantee of future popularity.

Expected Outcome

Trenvo aims to convert large and continuously changing music data into meaningful insights through rankings, charts, comparisons and predictions. The project provides users with a single, organized interface for discovering trending songs, evaluating artists and genres, monitoring changes over time and exploring potential future trends. Its modular structure separates data loading, preprocessing, analysis, prediction and UI components, making the system easier to maintain, extend and connect to live data sources.

Core Stack	Data Sources	Focus
Python • Pandas • NumPy • Scikit-learn • Streamlit	CSV / Spotify • YouTube • Search Trends • Social Metrics	Music analytics • Trend detection • Future trend prediction